

Surveyed 2026-07-25. Supersedes the 2026-07-23 version, which covered only the academic tier and predated the NeurIPS 2025 results.

Honest positioning: this is an active field and several groups are ahead of ABRA on agents. This document maps each public attempt against ABRA, states what we can take from it, and — because it determines whether we may take anything at all — records the licence.

1.1 The strongest public battling agents are RL and MCTS, not LLMs. The PokéAgent Challenge (NeurIPS 2025) drew 100+ teams and 650+ participants. Track 1, competitive battling, was won by **PA-Agent (RL)** and **Foul Play (MCTS)**; the LLM entry (August) took a Judges' Choice rather than the win. The organisers' summary is blunt — *RL specialists dominated partially observable two-player games*. The paper adds that Pokémon battling is "nearly orthogonal to standard LLM benchmarks."

Consequence for ABRA: the LLM route is not the play for decisions. Keep LLMs where PokéLLMon showed they earn their place — explanation and knowledge retrieval, i.e. KADABRA's prose — and make the decisions with search and learned policies.

1.2 A public VGC doubles corpus EXISTS, in our exact regulation, under MIT.

This is the correction that matters most, and it reverses an earlier draft of this document.

[cameronangliss/vgc-battle-logs](#) — **MIT licensed**, four files whose names are Showdown format ids:

logs_gen9championsvgc2026regma.json	47 MB	
logs_gen9championsvgc2026regmabo3.json		
logs_gen9championsvgc2026regmb.json	2 MB	<- ABRA's exact format
logs_gen9championsvgc2026regmbbo3.json	26 MB	

gen9championsvgc2026regmb is character-for-character the id in `engine/champions_sim.js`. It is Pokémon Champions, not Scarlet/Violet — the paper's "Scarlet and Violet" phrasing refers to the underlying Gen 9 species data.

MEASURED, because the 88,905 headline is misleading for our purposes. Downloaded and counted:

File	Games	Dates
Reg M-B Bo1	324	2026-06-18 → 06-20
Reg M-B Bo3	3,843	2026-06-17 → 06-20

The 88,905 total is dominated by Reg M-A and the Bo3 formats. **In ABRA's exact format the dataset holds 324 games against our 8,356 — we have 25× more.** Id sets are disjoint (theirs 2635..., ours 2653...): zero overlap, different weeks.

Collection also **stopped on 2026-06-20**, over a month before this survey. ABRA's hourly ingest is live. So on volume and recency in Reg M-B Bo1, ABRA is ahead, not behind.

What is genuinely worth having is not games, it is SETS. Every file is filtered for **Open Team Sheets**, so the full set is declared. Across the two M-B files that is roughly **50,000 complete sets** in our format. ABRA has almost none: a closed-sheet replay reveals a mean of 1.38 of four moves, 69.7% no item, 75.5% no ability, and ADR-001 records twice that *whatever fills that gap dominates any result*.

`engine/set_priors.js` fills it from usage priors and has **never been checked against a real set, because we had none to check against.**

Consequence for ABRA — three uses, in order of value:

1. **Ground truth for set reconstruction.** `engine/set_priors.js` currently guesses the missing 2.6 moves from usage priors and has never been checked against a real set, because we had none. OTS games give real complete sets to validate against. This converts our biggest unquantified assumption into a measurable error rate.
2. **10× the data**, in-format, for anything that does not depend on hidden information — matchup structure, team construction, damage, policy training.
3. **Raw Showdown logs**, so `extract()` parses them with no new code.

The honest caveat: OTS is a **different information regime** from ABRA's closed-sheet Bo1 ladder. XATU's entire subject is what is hidden and how belief narrows on reveal — a question that does not exist when both sheets are open. So this corpus supplements the ladder store for engine and policy work; it cannot replace it for belief work, and the two must not be pooled without saying which is which.

Foul Play, the Track 1 co-winner, is still singles-only and cannot play our format. But "nobody has a doubles agent" was wrong: VGC-Bench's PSRO and BC agents play doubles, and they are cross-evaluated. ABRA is behind on agents in doubles, not ahead.

The current centre of gravity. Two tracks (competitive battling, RPG speedrunning), a live leaderboard, and — the part that matters most to us — **a released dataset of 20M+ battle trajectories** with heuristic, RL and LLM baselines.

FORMAT, CHECKED — and it changes the recommendation. Track 1 runs **Gen 1 OU and Gen 9 OU**, both **singles**. The 20M is 4M human replays plus 18M self-play, spanning generations but **not doubles or VGC**. The paper carries **CC BY-NC-SA 4.0**; the dataset's own terms are not stated in the paper and must be read from the HuggingFace repo before any use. NC would matter if ABRA ever becomes commercial.

What ABRA can take: less than it first appears. This is not a drop-in fix for our sample size, because it is the wrong format. It is a plausible **pre-training** corpus — learn Pokémon-general representations from singles, fine-tune on Reg M-B doubles — but singles→doubles transfer is unproven and doubles adds a partner, spread damage, redirection and a 4-of-6 bring. Treat it as an experiment with an acceptance bar, not as a solution.

Track 1 co-winner. Root-parallelised **Monte Carlo Tree Search** over `poke-engine`, a bespoke Rust battle engine, guided by a **hand-written evaluation function rather than rollouts**.

That last detail is the interesting one. Foul Play does not roll out to terminal; it searches and evaluates at the leaf — the DeepStack/AlphaBeta shape, and precisely the architecture SLOWKING and ALAKAZAM are specified to have. Someone has now won a competition with it in this domain.

What ABRA can take: the architectural confirmation, and the design lesson that a good leaf evaluator beats deep rollouts. ABRA already has the leaf evaluator — PORY.

LICENCE WARNING. Foul Play is **GPL-3.0**, which is viral. ABRA is MIT. Copying Foul Play code, or linking `poke-engine`, would force ABRA to become GPL. **Read it for ideas; do not vendor it.** This is the one project in this survey with a licence that constrains us.

The closest neighbour: **doubles VGC**, on poke-env, with four PSRO variants (pure self-play, fictitious play, double oracle, exploitation), behaviour cloning, an LLM wrapper, and three poke-env heuristics. Ships a scraping pipeline rather than logs. Cross-evaluated across 1 / 4 / 16 / 64 team settings.

Their headline result, and it changes MEW's priority: the strongest agent is **BCSP — self-play initialised with behaviour cloning**. Not pure self-play, not pure imitation. Clone first, then improve by self-play.

What ABRA can take: three things, and it is MIT so we actually may.

1. **The BCSP sequence.** MEW currently runs random self-play. Random is correct for plumbing and for matchup structure, but VGC-Bench says it is not what wins. The behaviour-cloned policy is the deliverable, not the upgrade.
2. **The PSRO implementations** — DITTO's rebuild has been "to do" for two versions and is implemented here, in our format, under a licence we can use.
3. **Their BC pipeline**, which turns logs into state-action trajectories. That is SLOWKING Paper 1 / `engine/game-spec.js`.

Their known weakness, which is ours to avoid: the LLM baseline is evaluated on **20 battles** against 200 for every other agent. That comparison cannot support a conclusion — the same underpowered-claim problem ABRA gated against on 2026-07-25 (`engine/eval_harness.py`).

Their open problem: as team diversity grows, the best single-team agent becomes more exploitable and generalises worse. Team space is $\sim 10^{139}$. Exploitability vs generalisation is unsolved.

Offline RL with transformers on **5M human trajectories + 20M self-play**, reaching the top decile of human players in singles, model-free, **with no search at all**.

What ABRA can take: the mixture ratio. Roughly 4:1 self-play to human. And the sobering lesson that a large enough learned policy may not need search — which means ALAKAZAM's search layer must be measured against the no-search policy and kept only if it earns its cost.

Expert-level minimax agent over the open engine, ~ 1500 Elo, no learned dynamics, no equilibrium guarantee. A domain proof and a baseline.

First LLM agent at human parity in randoms, via state→text prompting and retrieval-augmented generation. Superseded on strength by RL, but its RAG-of-knowledge layer is the right model for KADABRA's explanations.

The infrastructure layer. Python library for scripted agents, self-play and RL against a local Showdown server; subclass `Player`, override `choose_move`. Everything above is built on it.

What ABRA can take: it is the standard harness and it is MIT. ABRA currently drives the simulator directly through `BattleStream` in `engine/champions_sim.js`, which works and is verified. poke-env would matter if we want to reuse VGC-Bench's agents, since those are written against it.

Open question: whether poke-env's doubles support is complete for our format. VGC-Bench is built on it and does doubles, which is suggestive but not confirmation.

A decade of open-source Showdown bots that solved practical problems before the academics arrived: **Technical Machine** (David Stone), **pmariglia/showdown** (the ancestor of Foul Play), **MariBot**, and forks such as **jfiacco/showdown** and **Agetian/showdown-battlebot**. **PsyMew** is a recent Foul Play fork that swaps the decision engine for an LLM pipeline (Gemini or Claude).

What ABRA can take: mostly operational rather than algorithmic — protocol handling, reconnection, team validation, ladder etiquette. Technical Machine's deliberate non-determinism (it varies between good options so opponents cannot read it) is a small, real idea that anticipates the mixed-strategy argument by years.

Capability	Best public work	ABRA	Verdict
Singles battling agent	Foul Play (MCTS), PA-Agent (RL)	none	Behind. Not our format
Doubles/VGC agent	VGC-Bench (benchmark)	ALAKAZAM (spec only)	Both early. Genuinely open
Battle engine correctness	poke-engine (Foul Play, GPL)	official Champions mod, verified 31/31 vs @smogon/calc	Level, and ours is the authority
Data volume, Reg M-B Bo1	vgc-battle-logs: 324 games, stopped 2026-06-20	8,356 , live hourly	Ahead, 25×. No overlap
Data volume, Reg M-B Bo3	3,843 open-sheet games	~0, bucket scaffolded and empty	Behind. Worth adopting
Complete sets (ground truth)	~ 50,000 from open team sheets	~0; 1.38 of 4 moves revealed	Behind, and this is the one that matters
Belief-state / equilibrium search	Ihara et al. 2018 compared determinization vs information sets for Pokémon and found IS-MCTS better; belief-MCTS is a mature literature	SLOWKING/ALAKAZAM (spec, unbuilt)	Not first. Not even close. The narrow claim that survives is PBS re-solving in <i>doubles</i> , and that is a gap of degree, not of kind
Mid-game value function	implicit in RL agents	PORY — RETRACTED: ties a two-feature material baseline; see below	Comparable, and ours is explicit and measured
Evaluation honesty	weak — VGC-Bench's LLM at n=20	proper scores, CIs, baselines, power gate	Ahead. This is the real differentiator
Team building	VGC-Bench PSRO	DITTO (needs rebuild)	Behind. Adopt their PSRO

Adopt (MIT, no obstacle):

- VGC-Bench's PSRO implementations for the DITTO rebuild.
- VGC-Bench's BC pipeline instead of writing `game-spec.js` from scratch.
- poke-env if and when we need their agents to run.

Do NOT treat as a data fix:

- PokéAgent's 20M and Metamon's 5.3M are **singles only**. Neither closes ABRA's power problem, because neither is the format we study. Useful as a pre-training experiment; not a substitute for collecting Reg M-B doubles.

Read but do not vendor (GPL-3.0):

- Foul Play and poke-engine. Ideas only. Vendors would relicense ABRA.

Keep building — nobody else is:

- Belief-state search in doubles VGC.
- The measured-evaluation discipline. On the evidence of this survey it is ABRA's strongest and least contested claim.

Stop building:

- Anything that duplicates a scraper, a harness or a PSRO loop that exists under MIT.

Project	Licence	May we use the code?
VGC-Bench	MIT	Yes
poke-env	MIT	Yes
Foul Play / poke-engine	GPL-3.0	No — viral, ABRA is MIT. Ideas only
PokéAgent dataset	paper CC BY-NC-SA 4.0, dataset terms unstated	Read HuggingFace terms first; NC would block commercial use
Metamon	MIT (code); dataset terms unstated	Code yes. Data is singles only, terms unread
Pokémon Showdown / champions mod	MIT — verified from LICENSE, "The MIT License (MIT)"	Yes. Already used, pinned at 20ad99f . Vendoring data/mods/champions/ is permitted provided the copyright notice travels with it (ADR-001 migration step 1)
@smogon/calc	MIT	Yes. Already a dependency; it is the damage ground truth

All of the above were read from the LICENSE file itself, not from a badge or an abstract.

ABRA is MIT. Any GPL code entering this repository would change that for the whole project, which is a decision to be made deliberately and not by accident in a pull request. In practice the rule is simple: **everything ABRA actually depends on is MIT, and the single GPL project in the field (Foul Play / poke-engine) is one we only read.**

Checked 2026-07-25.

NeurIPS 2026 runs 6–12 December in Sydney, with satellites in Atlanta and Paris.

Route	Deadline	Status
Propose a competition	15 May 2026 (accept 15 June)	Closed
Propose a workshop	6 June 2026	Closed
Enter an accepted 2026 competition	varies	Open in principle — the accepted list is not published in the newsletter and must be read from the official competition page
MLRC 2026 → TMLR	30 September 2026	Open
arXiv preprint	none	Always open

The catch, and it is the important part: the PokéAgent Challenge is not VGC. Its battling track is Gen 1 OU and Gen 9 OU — singles. ABRA cannot enter its doubles work into a singles competition, and converting ABRA to singles would discard the one thing that makes it distinct.

So the realistic routes are (a) a paper rather than a competition, and (b) proposing a **VGC doubles** competition for NeurIPS 2027, where the pitch writes itself: the 2025 battling winner cannot play the format, and no public doubles corpus exists.

How the strongest engines in *other* games generate data and evaluate positions — Stockfish NNUE (billions of positions at modest depth), Leela Chess Zero (self-play with policy temperature ≈ 2.25 for deliberate diversity), KataGo (sample efficiency as a first-class target), AlphaZero (and why its no-metagame assumption does not transfer) — is treated in docs/MEW-whitepaper.md §2, where it informs MEW's design directly.

Three claims in an earlier draft of this document were wrong, and they were all wrong in the flattering direction. Recorded here rather than quietly edited out:

Claimed	Actually
"No public VGC doubles corpus exists"	One exists, MIT, with open team sheets — but 324 games in our exact format against our 8,356 , and it stopped collecting on 2026-06-20. The 88,905 headline is mostly Reg M-A and Bo3. Claimed "far behind on data" twice; wrong both times
"Nobody has done belief search in Pokémon"	Ihara et al. 2018 compared determinization vs information sets and found IS-MCTS better
"Nobody has a doubles agent"	VGC-Bench's PSRO and BC agents play doubles and are cross-evaluated

The honest position: ABRA is **behind on agents**, has **far less data than it should** given a larger in-format corpus is freely available, and is **not first** on belief search.

What is actually ours, stated plainly and without claiming novelty it does not have:

- a **closed-sheet Bo1** store, which is a different information regime from the public OTS corpus,
- an engine wiring **verified** against an independent damage oracle (31/31 within 2%),
- an explicit, calibrated mid-game value function (PORY — **retracted**: it ties `alive_diff + hp_diff` exactly, so it is material arithmetic, not a value net),
- and an evaluation discipline — proper scores, clustered CIs, honest baselines, and now a power gate — that this survey found the field does not consistently apply.

That is a sound engineering position. It is not a breakthrough, and the documents should stop implying otherwise.

- PokéAgent Challenge — [arXiv:2603.15563](#) · [pokeagent.github.io](#) · [results summary](#)
- Foul Play — [github.com/pmariglia/foul-play](#) (GPL-3.0) · [design notes](#)
- VGC-Bench — [github.com/cameronangliss/VGC-Bench](#) (MIT) · [arXiv:2506.10326](#)
- Metamon — [arXiv:2504.04395](#)
- PokéLLMon — [arXiv:2402.01118](#)

- poke-env — github.com/hsahovic/poke-env (MIT)
- PsyMew — [Smogon thread](#)